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Old English morphological analysis tool

Ondrej Tichý (Prague)

MORPHOLOGICAL ANALYSER OF OLD ENGLISH

The paper describes the construction and testing of an electronic application for semi-automatic morphological analysis of Old English.

It introduces the state of the art in the field of electronic analysis of Old English, provides aÿbrief overview of Old English morphology and discusses the reasoning behind our theoretical framework. An account of the chosen methodology is offered and aÿspecific description of its implementation is provided: from the acquisition and preparation of the lexical input data, through the programming of the forms generator to the testing of the results by analysing Old English text. The resulting recall of 95% is aÿsuccess; however, the paper also hints at how it may be improved. It also discusses further use and development of the analyser, especially the disambiguation of its results.

The paper makes aÿfuture semi-automatic morphological tagging of Old English texts aÿreal possibility.

KEYWORDS

Old English, historical linguistics, computational linguistics, automatic morphological analysis, morphology, forms generator

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1. INTRODUCTION1

Contemporary diachronic language research is increasingly turning to language corpora as a fundamental source of new knowledge and for verifying older knowledge. However, working with corpora of older languages has its own significant specifics and pitfalls. From the perspective of synchronous research, diachronic corpora are still not very extensive (corpora of up to several million positions/tokens are common) and, above all, they are only very sparingly marked. In synchronous corpora (especially modern English, but also Czech), we now commonly expect, for example, lemmatization or directly morphological and syntactic marking. We rarely encounter it in diachronic corpora. However, this fundamentally complicates the use of corpora for the analysis of historical language, because, for example, without lemmatization, it is practically impossible to perform collocational analysis and most frequency analyses (the basic method of corpus analysis) are much more demanding on unlemmatized material.

1 The article is a summary of an unpublished doctoral thesis of the same name defended in 2014 at FF UK.

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One of the main reasons for this state of affairs is that tagging (and for these purposes we will consider lemmatization as a type of morphological tagging) of a historical language is significantly more difficult than tagging of a contemporary language. This is mainly related to the possibilities of automating the tagging process. For modern English, there are already relatively reliable tools for morphological and syntactic tagging, after all, lemmatizing a modern English text is, for example, almost banal from the perspective of modern Czech, with some exaggeration. However, even for the undoubtedly more complex Czech grammatical morphology, there are high-quality automatic analysis tools. However, the problem does not lie only in morphological complexity, but primarily in standardization.

Anyone who has ever worked with a historical language or a non-standardized language variety knows that the spelling, forms, and rules of manuals often do not exactly correspond to the language of the documents. Just as language manuals of modern standard varieties (e.g., standard Czech) are abstractions over a more complex linguistic reality (langue/pa-role), manuals concerning historical varieties of the language are, in addition, abstractions over the complexity of dialects, scribal traditions, and idiosyncrasies or diachronic differences within the periods described.

The motivation for creating a tool for automatic morphological analysis of Old English is therefore primarily to facilitate or, in some cases, even enable corpus research of this period of the English language. As follows from the above, this tool must be able to cope with both the morphological complexity of Old English and the low standardization of its documents.

2. THEORETICAL FRAMEWORK

A closer examination of the previous results in the field of automatic processing of Old English morphology has shown that this is not a completely new problem, although it is undoubtedly a problem that has not yet been satisfactorily solved. Some of the previous attempts to automate morphological analysis have been limited to only part of the language structure. For example, *Verbix* (Lindberg, 1995), *The Nerthus Project* (Martín Arista, 2004) or *Morphological Analyzer for Old English Verbs* (Adams, 2007) focused only on verb morphology. Some of them dealt only with a narrow selection of material intended for analysis — e.g. *Morphological Analyser of Old English Texts* (Calle Martín et al., 1997–2001). Two projects that are not yet completed are directly linked to our database (or the same source of lexical data): the lemmatization of the Old English poem *Daniel* (Kleinman, 2012) and the lemmatization project of the *YCOE* corpus.2

A common feature of all these projects is a theoretical framework that determines the basis of the tool, a dictionary generated in various ways, with which the documents, i.e. the forms themselves occurring in Old English texts, are subsequently compared.
However, the method of comparison and subsequent analysis differs. In the simplest case (Kleinman), it is a simple comparison of character strings using the so-called Levenshteyn distance (a measure of similarity between text strings), in more sophisticated tools, the items are

2 *The Yorkÿ Torontoÿ Helsinki Parsed Corpus of Old English Prose.* Information about the Ans Van Kemenade and Erwin Komen project was obtained in 2013 through private correspondence.

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The dictionary (lemmas) are edited based on morphological rules and then compared with the documents.

We decided to proceed in a similar way in our case, but since the above-mentioned tools, or rather their description, are either unavailable or incomplete, we decided to draw inspiration in general terms from the implementation-verified solution of morphological analysis of modern Czech (Osolsobÿ, 1996; Sedláÿek, 1999; Sedláÿek — Smrž, 2001).

The theoretical framework established in this way assumes three basic phases of the solution:

1. **Preparation of data** that will form the tool's **core dictionary .** The core dictionary should be as exhaustive as possible in terms of Old English lexicon coverage and should contain, in addition to the lexicon itself, basic morphological information about individual items.

2. Creation **of a generator** that generates all grammatical forms of Old English words from the basic dictionary entries based on morphological **rules** and assigns the appropriate morphological categories to each form, i.e. **a dictionary of forms.**
3. Creating an application that will compare Old English text documents with entries in the form dictionary, i.e. **the analyzer** itself in the narrow sense of the word.
   In case of a match, the document will be marked with both a reference to the relevant entry in the basic dictionary (lemmatization) and to the morphological categories belonging to the matching entry in the form dictionary (morphological analysis).

However, for the resulting lemmatization and morphological tagging to be usable by the researcher, the results of the automatic analysis will still need to be disambiguated (either manually or partly by machine), since it cannot be assumed that the tool would be able to handle, for example, the selection of homonym forms based on morphology alone. However, the disambiguation process is no longer part of this morphological analysis tool.

As mentioned in the introduction, the specific implementation of all three phases necessarily stems from the specifics of Old English morphology and must also correspond to the actual state of the analyzed documents.

3. MORPHOLOGY OF OLD ENGLISH

Old English, i.e. the English from around 700–1100, is, in contrast to Modern English, a typologically still predominantly inflectional language, albeit with a considerably limited inflection compared to some Indo-European languages, such as Czech. In Old English we encounter a similar declension of nouns, adjectives and pronouns to that which we know, for example, from Modern German, and, just like in Modern German, verb conjugation is particularly developed in Old English.

The scope of this article does not allow for a detailed description of Old English morphology; we will limit ourselves to a few general remarks relating specifically to our tool.

The fundamental difference between traditional philological descriptions of Old English morphology (e.g. Wright — Wright, 1914), or modern descriptions intended for students (e.g. Baker, 2012), and the description resulting from the analysis of Old English morphology

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for the purpose of machine processing, is primarily at the level of detail that is devoted to various linguistic structures. Traditional descriptions of Old English morphology pursue other goals in addition to the description itself, e.g. an emphasis on the historical continuity of structures and phenomena, and sometimes distinguish in the description even structures that are no longer formally different in a given period (e.g. paradigms following paradigms that were originally different in their reconstructed form, but no longer formally different in the Old English material). Our description was, on the contrary, thoroughly pragmatic in this respect, and therefore, for example, did not distinguish even those structures whose formal difference according to the traditional description was lower than the actual formal variation of these structures in the text, whether it is a scribal, dialectal or diachronic variation.3

However, our description is in some respects more detailed, both in relation to the significantly truncated descriptions of modern students, and often also in relation to the traditional description — it must cover all the structures that the usual descriptions include as exceptions or neglect due to their marginality. In contrast to the traditional seven strong, three weak and a few preterite-present and irregular verb paradigms, our description introduces 132 verb classes for the purposes of machine processing.

Another difference that distinguishes our description from traditional ones is our orientation to the dictionary, and therefore to the lemma as the basic form of inflected forms, even in cases where the traditional and more philologically meaningful analysis is based, for example, on the stem or root. The reason is again pragmatic — preparing a basic dictionary of lemmas is significantly easier than preparing a dictionary of roots or form-forming stems. The principle of our description is therefore that using the morphological rules we have established, it must be possible to derive all inflected forms of a given word from any lemma. In the case of declension types, all inflected forms can usually be created only on the basis of the lemma and endings, while in the case of verb conjugations the situation is more complicated and will be described below in the section on implementation.

However, it is necessary to emphasize that at its core, our description is based on traditional descriptions, and unless there is a good reason not to, we try to stick to them. We assume that this will also make the resulting morphological analysis well understandable and transparent for users familiar with traditional descriptions.

4. IMPLEMENTATION

As we indicated in the section describing the theoretical framework of our solution, the implementation is divided into three basic phases.

4.1 DATA PREPARATION

Based on an exhaustive review of lexicographical sources mapping Old English material (Tichý, 2007), we decided to use lemmas for basic

3 It is therefore essentially a signal-to-noise ratio. If, for example, the traditional description distinguishes structures based on the formal difference *-on* vs. *-an,* but the actual occurrence of these forms is statistically random with respect to the traditionally described structures, we neglect this difference in our description.

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dictionary *An Anglo-Saxon Dictionary* (Bosworth — Toller, 1921, hereinafter referred to as BT). Its main advantage is the breadth of coverage, as it is the only dictionary to date that at least theoretically covers the entire breadth of the Old English lexicon. In addition, when preparing the data, it was possible to base it on a previous digitization project (Tichý, 2007), but it was still necessary to make a number of adjustments to the electronic version of the dictionary (marking morphological information, corrections to the macrostructure of the dictionary, etc.) before it was possible to export a list of lemmas with basic morphological information and identifiers allowing for referencing back to the original dictionary entries.

The basic dictionary data is further enriched with information *from Old English Grammar* (Wright — Wright, 1914), or enriched with morphological information are the lemmas that occur in both sources (see *Table* 1). The Wrights' dictionary (hereinafter referred to as the grammar) was used both because of its structure — all example words are included in the index in the form of lemmas and clearly refer to the relevant morphological information, and because it is a detailed, traditional and still widely used grammar. In contrast to the more detailed and modern Campbell's grammar (1983), it is also unencumbered by licensing and the extracted information could therefore also be used in the freely accessible online version of the Anglo-Saxon Dictionary (see bosworthtoller.com).

Finally, the basic dictionary was enriched with manually compiled morphological data concerning example verb paradigms, which enable the generator described in the next phase to take into account inflected forms as well as infixation (see *Table* 2).

4.2 GENERATOR

The form generator first loads the base dictionary data and performs basic standardization. This is a standardization that either has no impact on the analysis itself or its results (e.g., unification of the allographs *thorn and edh)* or gives rise to alternative forms of lemmas that will play different roles in the subsequent process (e.g., marking vowel lengths).4 When loading the data, the generator also automatically calculates some information about individual items, which is subsequently used for generating and assigning forms (e.g., number of syllables, stem length distinction, etc.).

There are two stages, in which the generator first assigns sample paradigms5 to the items of the basic dictionary and then generates individual inflected word forms based on the paradigm and the specific item.

The assignment of paradigms is carried out according to the following algorithm:

1. First, the entries of the basic dictionary are compared as text strings with the lemmas of the model paradigms, taking into account also the additional mor-

4 Although length marking reflects a real phonological difference in quantity, it does not occur in the original texts and is only a contribution of modern editors (e.g. BT or grammar authors). The generator and the analyzer therefore neglect it, but the alternative form of the lemma with diacritics taken from the original sources of the basic dictionary is preserved and is therefore available to users who, for example, wish to analyze a non-authentic modernly edited text, or who want to use the generated material for teaching and thus prefer the text with diacritics.

5 Exemplar paradigms are derived from grammar.

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ID	lemma	wright	noun	pron.	adj.	verb	part.	adv.	prep.	conj.	interj.	num.	w. vb.	s. vb.	c. vb.	pp. vb.	a. vb.	un. vb.	m. n.	f. n.	n. n.	u. n.
008031	DRÝ	142;388	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
008032	drý-cræft	NULL	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
008033	drý-cræftig	NULL	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
008035	drýfan	NULL	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
008036	drygan	530	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
008040	dryht	390	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
008041	Dryht	NULL	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
008042	dryht-bearn	NULL	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
008043	dryht-cwén	NULL	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
008044	dryht-ealdor	NULL	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
008045	dryhten	288;340;563	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
008047	dryhten-bealo	NULL	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
008048	dryhten-dóm	NULL	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

Table 1. Sample basic dictionary data

IDverb	type	class	subdivision	subc.	SN	variant	paraID	prefix	PreV	PoV	bnd	dent	end
2 tailor	s	[illegible]							[illegible]				an
2 styles 2	s	1 0 c 491 0 PaInSg1 1 0 c 491 0							sn á 0 ð sn i 0				0
styles	s	PaInPl							d				on
2 sníðan 1 0 c 491 0 Pp 74 fremman w 1 A b 526 0 Inf									sn i 0 d fr fr				in
74 fremman w 1 A b 526 0 PsInSg2 74 fremman w 1 A b									a mm				an
526 0 PaInSg1 74 fremman w 1 A b 526 0 PaInSg1 74 fremman									em				East
w 1 A b 526 1 Inf								fr	em			ed	and
								fr	em			from	and
								fr	em				an
74 fremman w 1 A b 526 2 Inf								fr	e m i				an

Table 2. Example of the structure of verb paradigms (verbs *sníðan and fremman)*

phological information (part of speech, verb type, noun gender, etc.) to avoid incorrect assignment of homonyms. The word-forming structure of lemmas is also taken into account — if one of the prefixes contained in BT is found at the beginning of the lemma, a match with the prefix is first searched for, then without the prefix. In the case of items that have not yet been assigned, the generator will try to find a match with at least part of the lemma of already assigned items, which will result in the assignment of a series of composites that are usually inflected similarly to their governing member.

2. The previously unassigned items are then assigned based on references to the model paradigms in the grammar. Since this increases the range of assigned items, the assignment of the previously unassigned items to these newly assigned items is repeated by comparing text strings according to the first step.

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3. This is followed by a morphophonological analysis6 of the previously unassigned items and their assignment to paradigms based on it. In this way, all previously unassigned adjectives can be assigned.
4. The remaining unassigned items are assigned on the basis of probability, i.e. to the most common paradigms corresponding to their morphology.
   a. All strong verbs are assigned to the *helpan pattern,* all others to the type *of daemon.*
   b. All masculine and indefinite gender nouns are assigned to the *stán pattern,* all feminines to the *ár type,* all neuters with a long stem to the *word* type , and all neuters with a short stem to the *hof type.*

The generation of inflected forms proceeds gradually by individual parts of speech, using concordances in inflection between parts of speech, so that, for example, inflected verbal participles are first generated with verbs in the basic form and then inflected with adjectives. Except for verbs, this usually involves simply removing any lemma ending, changing the morpheme seam, and adding the ending according to the given paradigm. In addition, information about the change of the vowel in the root syllable (infix) is used for verbs, so the structure of the lemma is first analyzed, the root vowel is found, and replaced either with a vowel corresponding to the paradigm (ablaut vowels) or with an umlaut vowel, which is derived from the original vowel according to the umlaut rules.

During generation, some forms are created in multiple alternatives, either due to doublets given by morphological or phonological rules or to frequent cases of orthographic, dialectal or diachronic fluctuations of forms. The resulting forms are also assigned information justifying the generation of the alternative form, so that, for example, forms from different dialects or scribal traditions can be distinguished.

Each generated form is also enriched with information about all relevant morphological categories, the sample paradigm, and the basic morphophonological structure on the basis of which it was composed.

Pronouns and some irregular verb forms would be so inefficient to generate (their number is small and the formal connection with the basic form is too small) that they are inflected manually for the purposes of the tool and thus inserted into the dictionary of forms.

An analyzer in the narrow sense is actually just an application that compares Old English text documents entered by the user with the forms produced by the generator and displays information about these forms that the generator assigned to them. However, a fundamental complication in assigning documents is caused by the aforementioned variation of forms. This is partly solved by the introduction of alternative forms during generation described above, but it is not completely solved, since many cases of variation are difficult to predict during the generation phase and, above all, if all possibilities of variation were to be taken into account during generation, the number of generated forms would grow unbearably (see below).

Therefore, the analyzer works with so-called variation filters, which allow it to assign even documents that do not correspond to any entry in the form dictionary without them. These filters are

6 Number of syllables, their length, etc.

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for war	verbs para-slov. para-digma	war	Slovak: paradigm	Slovak: paradigm	words.
sealfian	2342 hyngrian 23 hycgan 1758 sníþan		10 spanan 9	6 pÿcan	3
that day		22 hebban	þerscan	6 streccan	3
Nerian	387 bregdan 21 rÿdan		9 windows	6 ounces	3
bidan	226 don	20 rice	9 beats	5 agan 5	2
the napkin	200 health	20 sécan	9 locks	briothan	2
forward	176 real 19 wríþan 123 cuman 18		9 eggs	5 connan	2
drencan	libban		8 tellan	5 dreccan	2
end	115 cweþan 18 scéadan 112 twéogan		8 thousand	5 years 5	2
béodan	18 wegan 108 hléapan 15 willan 95		8 cars	years ago	2
helpful	wyrcan 15 brengan 81 gierwan 14		8 gilþan 7	4 dwell	2
methane	reccan 73 hátan 14 séoþan 14 stellan		hnigan	4 fricgan 4	2
weorþan			7 rÿcan	frignan 4	2
beran			7 slÿþan 7	gellan 4	2
weorþan	57 times		smugugan	migan	2
fealdan	55 give 53	13 steps 13	7 tÿcan	4 mold	2
jar	give 52 give	steps 13 steps	7 thin	4 nugan 4	2
teon	46 give		7 wÿcan	sceþþan 4	2
the blasphemy		12 béon	6 weccan	scippan	2
blue	40 men	12 little ones	6 the mix	3 steps 3 needy	2
set	39 swerian 12 féolan		6 cweccan		2
assign	30 stands	11 found	6 beehives	3 wleccans	2
ceosan	28 years old	11 lcan	6 degrees	3 beers	1
seon	27 good 11 bad		6 months	3 magan 6	1
lÿcan	26 licks	11 seal	míþan 3 nágan 6	murnan 3 nyllan	1
bid'an	23 seconds	11 psychos			1
phone	23 bands 10 strings		6 hours	3 scullion	1
				sporn	1

Table 3. Numbers of verbs assigned to individual paradigmatic patterns

based on multiple sources (primarily Baker, 2012, and Wright — Wright, 1914) and involve both simple character substitutions (e.g., the interchangeability of u and v) and complicated substitutions using regular expressions (e.g., the fluctuation of back vowels before nasals). The advantage of working with filters only at this stage is their selectability from the user's perspective — for example, the user can decide that, given the age and dialect of the text, he wants to use only some or none.

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ID	form_i lemma 240508 BT	stem	form
æt-swymman **240509** ætswumon 1014 æt-swymman 1014 æt-	ætswummon 1014	swimming pool	ætswummon
swymman 1014 æt-	swymman 1014 æt-swymman 1014 æt-	swimming pool	ætswummon
240510 ætswumme	swymman **1014** æt-swymman	swimming pool	ætswumme
240511 ætswume	240514 ætswummen 1014 æt-	swimming pool	ætswumme
240512 ætswumme	swymman 1014 æt-swymman	swimming pool	ætswumme
240513 ætswume	1014 æt-swymman 1014 æt-	swimming pool	ætswumme
swymman 1014 æt-swymman **240520** ætswummen 1014 æt-		swimming pool	ætswummen
240515 women	swymman **240521** ætswumen	swimming pool	ætswummen
240516 caustic fungus	1014 æt-swymman **240522**	swimming pool	caustic fungus
240517 ætswam	ætswimman 1014 æt-	swimming pool	caustic fungus
240518 caustic fungus	swymman **240523** ætswiman	swimming pool	caustic fungus
240519 ætswam	1014 æt-swimmer	swimming pool	caustic fungus
		swimming pool	ætswummen
		swimming pool	ætswummen
		swimming pool	swimming pool
		swimming pool	swimming pool

Table 4. Example of generated forms of the verb *æt-swymman*

The resulting analysis of the *andswarodon* document then looks like this:

ANDSWARODON

- ***and-swarian*** (infl. like *sealfian)*
  - **verb,** Pl. Ind. Pret. (w, 2, a, *and-sw-a-r-0-od-on)7*

4.3 RESULTS

The results of the tool were tested on excerpts from ten Old English texts totaling approximately 2,500 words. The individual excerpts were selected to ensure coverage of different dialects, genres, places and periods of origin (see *Table* 5).

As we have already stated in the section describing the theoretical framework, the purpose of the tool is only the analysis itself, not the disambiguation of the analysis results. For us, the measure of the success of the tool is therefore primarily recall *rather* than precision . Recall shows which parts of words/documents from the analyzed text were assigned correct morphological information by the analysis. However, it does not show how many alternative but incorrect solutions the analysis further assigned to individual documents. Although precision will be crucial for possible disambiguation, and not directly for our tool, the fewer alternative analyses our tool offers while maintaining the correct analysis, the easier the subsequent disambiguation will be.

7 It is therefore the indicative plural of the preterite of a weak verb of the 2nd class of subtype a, in the basic dictionary form *and-swarian,* conjugated according to the pattern of the verb *sealfian.*

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formParts	var	prob.	function	wright	pEx	pID	wclass	c1	c2	c3				
0-æt-sw-u-mm-0-on 0			painpl	Null	bindan	10	verb	painpl	null	PaInSg2	s	3	a	
0-æt-sw-u-mm-0-on 0 1			null	PaInSg2	null		bindan	10	verb	bindan	s	3	a	
æt-sw-u-mm-0-e	0		PaSuSg	null			10	verb	bindan	10 verb	s	3	a	
æt-sw-u-mm-0-e	0 1		PaSuSg	null							s	3	a	
æt-sw-u-mm-0-e	0		PaSuPl	null			bindan	10	verb	bindan	s	3	a	
æt-sw-u-mm-0-e	0 1						10	verb	bindan	10 verb	s	3	a	
æt-sw-u-mm-0-en	0										s	3	a	
æt-sw-u-mm-0-en	0 1		PaSuPl	null			bindan	10	verb	bindan	s	3	a	
0-æt-sw-a-mm-0-0 0			painsg1	null			10	verb	bindan	10 verb	s	3	a	
0-æt-sw-a-mm-0-0 0 1			painsg1	null							s	3	a	
æt-sw-a-mm-0-0	0		PaInSg3	null			bindan	10	verb	bindan	s	3	a	
æt-sw-a-mm-0-0	0 1		PaInSg3	null	null		10	verb			s	3	a	
0-æt-sw-u-mm-0-en 0			null		papt		bindan	10	verb	bindan	s	3	a	
0-æt-sw-u-mm-0-en 0 1			papt	if	null		10	verb	bindan	10 verb	s	3	a	
0-æt-sw-i-mm-0-an 0											s	3	a	
0-æt-sw-i-mm-0-an 0 1			if		null		bandage	10	verb		s	3	a	

The coverage in the tested excerpts was on average 95%, after switching on the variation filters it increased to 97.3%, i.e. only 2.7% of the texts were not assigned the correct morphological information by the analysis. As can be seen *from Table* 5, the best coverage of the tool is achieved in prose texts of West Saxon provenance from approximately the year 1000. Such a result corresponds to our expectations, because these texts are closest to the variety on which the grammar and to a large extent also the BT are based (although parts of the dictionary are oriented rather to the younger and poorly documented West Saxon period). On the other hand, the tool causes problems with non-prosaic and especially dialectal texts. Therefore, variation filters also respond best to dialects, as can be seen from the results for *the Rushworth Gospels.*

5. TECHNOLOGIES USED

Technologically, the first two phases are processed using scripts in the Perl language, which works with data in text files in Unicode encoding. It is assumed that these scripts are only run by the tool administrator in the event that there is a change in the source data (dictionary or rules). On the other hand, the third phase is processed in the form of a web application programmed in the PHP language and with data in the MySQL relational database, since it forms the environment for end users.

The user interface is not yet publicly accessible, but it is hosted together with the BT online dictionary on the server of the Faculty of Arts, Charles University.
Those interested in access can contact the author.

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name	description	genre	dialect	dating	positions	coverage	coverage with filters
House on the Rock (or the Parable of the Wise and the Foolish Builders)	West Saxon Gospels, Matthew 7:24–27	religious pro-for	990	West Saxon	100	100%	100%
Wise and Foolish Virgins (or the Parable of the Ten Virgins)	West Saxon Gospels, Matthew 25:1–13	religious pro-for	990	West Saxon	194	100%	100%
The Voyages of Ohthere and Wulfstan	from Histori-arum Advers-sum Paganos Book VII	travelogue in prose	890	West Saxon	347	99,4%	99,4%
Cynewulf and Cyneheard	entry for 754 AD in the Anglo-Saxon Chronicle (Parker MS)	chronicle in prose	890	West Saxon	313	99%	99%
Bede's „The Sun and the Moon"	from Ælfric's translation of Bede'sÿDe In times	professional prose	1000	West Saxon	146	98,6%	98,6%
Wulfstan's Sermon Wolves to the English	from Wulfstan's sermon	sermon in prose	1015	West Saxon (written in Northumbria)	320	97,1%	97,8%
Beowulf	lines 702–757	heroic poetry	that. 1000	mixture	313	96,8%	98,4%
Rushworth Gospels	interlinear gloss to Latin text of Matthew 6:1–6	religious pro-for	late 10c	Mercian	167	94.6%	97%
Lindisfarne Gospels	interlinear gloss to Latin text of Matthew 6:1–6	religious pro-for	10c	North-Umbrian	183	89,6%	90,7%
Oswulf's Charter charter		legal 806		Kent 471	84.7%	94.2%	
Total					2554	95%	97,3%

Table 5. Results of analysis on excerpts of Old English texts

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6. DISCUSSION AND CONCLUSION

As can be seen from Table 6, the total number of generated forms is high relative to the number of lemmas: ca. 71 thousand lemmas correspond to almost 12 million inflected forms. This fact is due to two factors.

From a practical point of view, there was no need to solve the problem feared by the authors of similar tools, i.e. the so-called over-generation. A great advantage of the material intended for our tool is that there is an exhaustive Old English corpus for Old English (Healey et al., 2000) and it cannot be expected that further significant evidence from this period would appear in the future. It is therefore possible to compare all generated forms with all evidence from the corpus and thus easily get rid of all unattested ("over-generated") forms. Therefore, we could afford to solve some of the exceptions or ambiguities of Old English morphology by simply generating other theoretically possible forms, without considering their possible evidence.

The second reason is the state of Old English morphology itself. As can be seen again from Table 6, the largest number of forms per lemma is shown by adjectives and adjectivally inflected participles, which is due to the fact that in Old English each adjective can be inflected according to at least two patterns (weak and strong) and in all three degrees. This dramatically increases the number of possible forms, although it is precisely adjectival morphology that is otherwise formally the weakest. Thus, out of the average 300 generated forms, there are on average only 47 truly formally distinct (unique) forms per adjective. These numbers clearly demonstrate the formal "weakness" of Old English adjectival morphology, which also corresponds to its very early demise.

The problem of re-generation is therefore particularly acute with regard to the number of attested but homonymous forms that will need to be disambiguated. On average, each analyzed document is assigned about 2 lemmas and several dozen possible morphological interpretations (the specific numbers vary according to parts of speech, see above).

Despite such a high number of generated forms, a small percentage (see Results) of documents remains undetermined or incorrectly determined. There are therefore also attested forms that the tool does not generate. Most of them are covered by variation filters, since generating all possible dialectal forms and forms corresponding to all period writing methods would be inefficient (regeneration would then be several orders of magnitude higher and the dictionary of forms would be difficult to use from a practical point of view). However, there are both unattested potential forms (i.e. grammatical forms in unlikely combinations of orthographic, diachronic and dialectal varieties), and attested forms that we would not mark as potential from the point of view of our tool, because they are unpredictable from the point of view of the dictionary, rules and described varieties (e.g. errors). Such forms will have to be marked manually in the disambiguation phase.

Disambiguation itself can be partly done mechanically (using syntactic rules or using training data and machine learning), and data assigned to alternative forms during generation can also be helpful in this process, which can indicate the probability that a given form will occur among documents. The more combinations of pre-generated variations and variation filters are used, the less likely the form is. Similarly, some morphological forms are considered less likely in a certain form based on manuals. However, a certain role will be played by

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part of speech	lemma	for	for / words	unique forms	uf / slov
adjective adverb	6 235	1 882 110	301,9	294 349	47,2
	1 863	16 640	8,9	15 652	8,4
	54	61	1,1	55	1,0
interjection conjunction	25	26	1,0	26	1,0
noun	19 835	259 496	13,1	140 837	7,1
masculine	8 668	93 874	10,8	59 251	6,8
feminine	7 306	122 508	16,8	57 133	7,8
neither	4 067	43 114	10,6	27 477	6,8
numeral	110	3 344	30,4	773	7,0
preposition	148	160	1,1	153	1,0
pronoun	43	4 996	116,2	1 010	23,5
verb	6 959	561 736	80,7	395 506	56,8
weak	5 094	461 239	90,5	325 768	64,0
strong	1 786	103 313	57,8	74 128	41,5
préteritopresentní	32	1 245	38,9	813	25,4
irregular participle	59	2 481	42,1	1 569	26,6
	35 642	8 715 027 244,5		1 247 483	35,0
last time	12 959	2 930 381	226,1	433 783	33,5
present	22 683	5 784 646	255,0	814 928	35,9
Total (avg. ratio)	**70 914**	**11 443 596 161,4**		**2 095 844**	**29,6**

Table 6. Numbers of generated forms by parts of speech compared to the numbers of lemmas

The manual work of experienced philologists may also have to be used in disambiguation, even if only because of morphologically unmarked or poorly marked documents.

Given the necessary and expected step of semi-automatic disambiguation, we consider the results of our tool to be successful, although we assume that this subsequent phase, necessary for high-quality marked text, will yield a number of insights that will help to further improve our tool.

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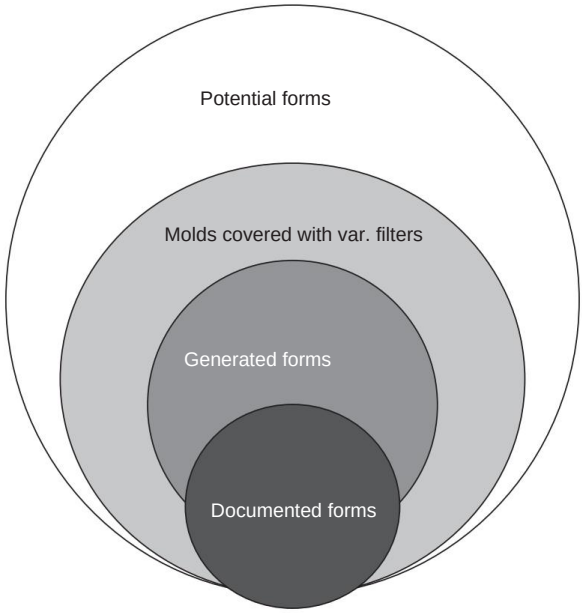

Figure 1. Overlay of the set of potential, generated, and documented forms

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Ondÿej Tichý | Department of English Language and Didactics, Faculty of Arts, Charles University | nám. Jana Palacha 2, 116 38 Prague 1
ondrej.tichy@ff.cuni.cz